Graph-Augmented Vision Transformers for Remote Sensing Image Scene Understanding

Dissertation working draft: Chapters 1–4
with a supplementary external-comparison chapter

English chapter draft

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This partial draft contains the introduction, literature review, method, experimental setup and supplementary external-method comparisons. Formal front matter, complete internal results and ablations, general discussion and conclusions remain to be integrated.

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Chapter 1

Introduction

1.1 Background

Remote sensing scene classification assigns semantic categories to aerial or satellite image patches. It provides a means of organising image archives and describing land use and land cover at the scene level. These descriptions can support applications such as urban analysis and environmental monitoring, although an image-level prediction does not locate every object or delineate every land-cover boundary. Surveys by Zhu et al. [1] and Cheng et al. [2] place scene classification within the wider development of learned representations for remote sensing.

The target description depends on the task. A single-label benchmark assigns one scene category to each image, even when several visual elements coexist. NWPU-RESISC45 and the Aerial Image Dataset (AID) provide examples of this setting [3, 4]. A multi-label task instead allows an image to carry several land-cover labels, as in BigEarthNet [5, 6]. This distinction changes the prediction objective and evaluation: choosing one class differs from ranking and thresholding multiple label probabilities. It also motivates examining regional representations in both settings, without assuming that a useful representation for one will transfer unchanged to the other.

Scene appearance presents a further challenge. Images within one category may differ in scale, orientation, texture and composition, while different categories can share visually similar components [3]. Consequently, recognising local elements and combining their contextual information are closely connected problems. An image representation must retain useful evidence while reducing the many spatial features to the compact vector required by a classifier.

Transfer learning offers a practical starting point. ImageNet established a large natural-image resource [7], and Nogueira et al. [8] examined how pretrained convolutional networks can be used for remote sensing scene classification. Domain-specific pretraining provides another route: Seasonal Contrast uses temporal imagery for self-supervised representation learning [9], while SatMAE adapts masked pretraining to temporal and multispectral satellite data [10]. These approaches illustrate that pretraining and downstream architecture are separate design choices. This thesis holds the recorded pretrained backbone source common within its internal comparison and concentrates on the structure added after feature extraction.

1.2 Motivation

The shared backbone is Swin-Tiny (Swin-T), a hierarchical vision Transformer whose shifted-window design enables interaction across neighbouring windows [11]. Its final tokens already contain contextual information. The motivation for adding a graph is therefore to investigate an additional regional inductive bias, rather than to remedy an assumed absence of spatial relationships in the backbone. Whether that bias is useful is an empirical question: an extra module can introduce helpful structure, redundant computation or additional sensitivity to training.

Regional aggregation offers an intermediate representation between dense feature tokens and a single global vector. Grouping tokens into a smaller set of nodes makes their interactions explicit. Graph attention can then aggregate neighbouring node features using learned weights [12]. However, this description leaves important choices unresolved. Region membership, pooling weights, edge selection and the return of graph features to the classifier each determine what information is preserved and exchanged. An apparent improvement from a complete architecture cannot automatically identify which choice caused it.

This thesis examines two grouping strategies within a Graph-Augmented Vision Transformer (GAViT). The fixed strategy uses a deterministic spatial partition and learns the relative importance of tokens within each region. The soft strategy constructs image-dependent prototypes and permits contributions to nearby anchors. Both retain a common anchor layout and explicit edge budget. The comparison asks whether adaptive membership improves on a simple spatial prior when the backbone and subsequent graph machinery are shared. It does not equate attention-weighted pooling with learned segmentation, or assume that soft assignment necessarily yields more meaningful regions.

The value of this structure also depends on its cost and consistency. Parameter count, latency and memory describe different aspects of the added computation. Repeated training can reveal sensitivity that a single favourable score conceals. Visual inspection can show actual assignments, connections and prediction changes, but cannot alone establish semantic correctness or causal benefit. These considerations motivate evaluating the complete accuracy–cost trade-off alongside diagnostic evidence.

1.3 Research questions

Three research questions organise the investigation:

1. **RQ1: Regional graph reasoning.** Does fixed-partition GAViT improve classifi-

cation relative to the shared Swin-T baseline, and how consistent is the observed difference across eligible training repetitions?

2. **RQ2: Adaptive membership.** Does local prototype soft assignment improve on fixed grouping under the same task protocol, and what changes can be observed in regional aggregation, graph connectivity and predictions?

3. **RQ3: Practical value and interpretation.** How do classification differences relate to additional computational cost, and what do successes, failures and visual diagnostics support about the role of the regional representation?

These questions permit conditional and negative findings. A more flexible grouping mechanism is not treated as preferable merely because it contains additional learnable parameters.

1.4 Contributions and scope

The work makes three contributions. First, it implements a Graph-Augmented Vision Transformer that combines attentive regional aggregation, a sparse spatial-and-feature graph, graph attention and region-to-token feedback. The architectural description distinguishes fixed membership from learned pooling and distinguishes feature-based edge selection from graph-attention weights. This makes the computational roles of the components explicit.

Second, it formulates an image-dependent local prototype grouping mechanism within the same architecture. Distinct normalisations define token membership and regional pooling, while feedback uses the membership weights. This allows the comparison to address adaptive aggregation without replacing the shared backbone. The soft-versus-fixed comparison concerns the resulting grouping pathway as a whole; it does not separately identify every effect of the added projection and feedback map.

Third, it establishes an evaluation framework combining single-label and multi-label classification, explicit model-selection rules, training repetitions, computational measurements and qualitative diagnostics. Calibration and final fitting are distinguished, and repeated training on a frozen partition is distinguished from repeated random data splitting. The contribution is the implemented design and its evidence-based examination; consistent superiority over the baseline is a proposition to be tested rather than an assumed outcome.

The empirical scope comprises red, green and blue (RGB) scene classification on NWPU-RESISC45, AID and BigEarthNet-19. BigEarthNet is restricted to three optical bands in this study, so the work does not evaluate full multispectral, radar or mul-

titemporal fusion. It also does not perform object detection or semantic segmentation. Regional visualisations describe internal feature aggregation, not independently validated object masks. Independent implementations of HGTNet and DFAGCN form a complementary comparison track. Chapter 5 reports the completed single-seed HGTNet measurements on NWPU-RESISC45 and AID, and DFAGCN on NWPU-RESISC45. They share the frozen data partitions and information-isolation rules with the internal models, while retaining separately disclosed training recipes. These supplementary results do not fulfil the remaining requirement for external comparisons under a unified protocol with multiple training seeds.

1.5 Organisation of the thesis

Following this introduction, Chapter 2 reviews remote sensing Transformers, image graph networks and regional grouping. Chapter 3 presents GAViT and its fixed and soft grouping variants. Chapter 4 specifies datasets, preprocessing, model selection and evaluation. The present partial draft adds Chapter 5 as a supplementary comparison of the completed external-method runs. The complete internal classification results and component studies, integrated discussion of cost and qualitative evidence, and conclusions remain to be incorporated into the final thesis structure.

Chapter 2

Literature Review

2.1 Scope and comparison perspective

Remote sensing scene classification connects visual representation learning with a question about spatial organisation: which arrangements of land-cover elements distinguish one scene from another? Cheng et al. [2] reviewed this problem through its methods, benchmarks and remaining challenges. Thapa et al. [13] subsequently examined deep-learning-based scene classification. These reviews provide a broad context for the present study, whose focus is the narrower relationship between contextual image features, regional aggregation and explicit graph reasoning.

This chapter follows three connected streams: visual backbones for remote sensing, graph-based image understanding, and region grouping. It compares methods along five dimensions: the visual unit represented by a node or token; the rule defining its neighbours; the operation aggregating their information; the position of the relational module; and the supervision used to learn the representation. This organisation separates architectural similarity from evidence that two methods solve the same task under comparable conditions.

Scene classification assigns an image-level description. A single-label benchmark selects one scene category, whereas a multi-label benchmark permits several land-cover labels for the same image. Neither task directly requires object boxes or semantic masks. Thus, a region useful for classification need not coincide with an annotated object. This distinction matters when grouping and graph visualisations are interpreted: class supervision can reward a predictive representation without enforcing a human-defined segmentation.

The chapter includes general vision methods where their mechanisms directly inform GAViT. Their inclusion does not imply that their natural-image, video, point-cloud or segmentation results transfer unchanged to remote sensing classification. Likewise, the review distinguishes a design precedent from a reproduced experimental comparator.

2.2 Learning contextual visual representations

2.2.1 Convolutional features and transfer to remote sensing

Convolutional neural networks (CNNs) learn spatially shared feature extractors whose receptive fields expand through successive layers. Simonyan and Zisserman [14] investigated deep networks built from small convolution filters. He et al. [15] introduced residual learning to facilitate optimisation of substantially deeper networks. These designs provide two relevant foundations: hierarchical features describe images at different levels, and residual paths allow an additional transformation to modify an existing representation. Neither property alone specifies an explicit graph between image regions.

The usefulness of natural-image pretraining for aerial scenes was investigated directly. Penatti et al. [16] examined whether deep features generalised from everyday objects to remote sensing and aerial scenes. Hu et al. [17] studied transferring pretrained CNN representations to high-resolution scene classification. Nogueira et al. [8] further investigated ways of exploiting CNNs for this task. Together, these studies motivate treating pretraining and adaptation as experimental factors, rather than attributing every performance difference to the last architectural module.

Scale handling is another example of a distinct design choice. Xie et al. [18] investigated a scale-free CNN that replaced fully connected processing with convolutional operations and global pooling to accommodate variable image sizes. Accommodating scale variation through the feature extractor differs from constructing content-adaptive region membership. These mechanisms may address related scene variation without being equivalent architectural interventions.

Transfer also affects the interpretation of a common-backbone comparison. Using the same pretrained feature extractor provides a controlled starting point, but joint fine-tuning can change that extractor differently in two models. A graph-augmented model therefore includes both its added relational operations and the representation learned in their presence. Removing its graph after training does not recover an independently trained backbone baseline. This is an experimental distinction, rather than a limitation specific to CNNs or Transformers.

CNNs remain a relevant comparison class. Liu et al. [19] revisited convolutional design and introduced ConvNeXt through a systematic modernisation of a residual network. This work cautions against explaining architectural performance solely by a CNN-versus-Transformer label. For the present study, the main backbone is held fixed so that the regional branch can be examined without simultaneously replacing the entire feature

hierarchy.

2.2.2 Attention, image tokens and hierarchical backbones

Vaswani et al. [20] introduced the Transformer around attention-based sequence processing. Its relevance here is the content-dependent combination of representations, rather than the original language task. Dosovitskiy et al. [21] applied a Transformer to image patch sequences in the Vision Transformer (ViT). Patch tokenisation supplied a regular input representation; attention subsequently contextualised those tokens. The resulting features should not be interpreted as independent unprocessed image blocks.

Training choices remained consequential. Touvron et al. [22] studied data-efficient image Transformers and attention-based distillation. Their work demonstrated that architecture, training recipe and teacher information must be considered together. A comparison using different pretraining data or distillation supervision does not isolate the effect of a regional graph, even when both models contain Transformer layers.

Hierarchical Transformers provided another route to organising spatial information. Wang et al. [23] proposed the Pyramid Vision Transformer, which progressively reduced spatial resolution to form a feature hierarchy. Liu et al. [11] used local windows and shifted-window connections in Swin. These approaches made multi-scale and cross-location interaction part of the backbone design. Consequently, adding a graph above Swin tests a further relational organisation of features that already contain context; it does not introduce context into an otherwise context-free model.

Backbone token resolution constrains the regional branch. A late-stage representation is computationally compact and carries high-level information, but its spatial units can mix several land-cover elements. Moving a graph to earlier features would change its node count, receptive fields and cost. Similarly, treating every patch as a node differs from aggregating several contextual tokens into one region. This motivates examining node construction and insertion depth alongside the choice of graph operator.

2.2.3 Remote sensing Transformers and domain pretraining

Bazi et al. [24] investigated vision Transformers for remote sensing image classification. Zhang et al. [25] also studied Transformer-based remote sensing scene representation. Their relevance is the application of token interactions to scene-level recognition; using attention does not by itself imply that the model learns explicit geographic objects or their semantic relationships.

Xu et al. [26] more recently combined high-frequency-enhancing Transformer processing with multi-layer contextual learning for remote sensing scene classification. This

provides a further example of treating feature detail and contextual fusion jointly. Its frequency-based design differs from defining a small graph of pooled regions, and its reported results do not identify which mechanism would improve the shared-backbone protocol used here.

Recent representation learning also changes what is available before task fine-tuning. He et al. [27] proposed masked autoencoders, using an asymmetric encoder–decoder to reconstruct missing image content. In remote sensing, Cong et al. [10] considered temporal and multispectral information in masked-autoencoder pretraining. Reed et al. [28] incorporated scale awareness, while Noman et al. [29] investigated multi-scale reconstruction. These works make the pretraining signal and spatial scale important parts of the representation-learning problem.

Liu et al. [30] developed RemoteCLIP through visual–language representation learning for remote sensing. Such supervision differs from learning only from the image-level labels of a downstream scene dataset. Additional text, temporal observations or spectral bands can improve the information available to a model, but they also change the comparison conditions. This thesis uses RGB inputs and a common recorded pretrained backbone in its internal architectural comparisons. Domain-pretrained methods are therefore relevant context, rather than automatically interchangeable baselines under that protocol.

Across these studies, richer pretraining and explicit relational structure address different design choices. Stronger initial features may reduce the remaining benefit of an added module, or may make meaningful grouping easier. Neither possibility follows from architectural inspection alone. Holding the backbone initialisation and downstream data protocol fixed is useful for testing one specific regional design, while broader backbone comparisons would answer a separate question about representation quality.

2.3 Explicit graphs for visual understanding

2.3.1 Nodes, adjacency and message aggregation

A graph representation requires a definition of its entities and connections. For an image, nodes may describe pixels, patches, pooled regions or objects. Spatial adjacency, feature similarity and known semantic relations encode different assumptions about which nodes should exchange information. A network that predicts scene labels from a feature-neighbour graph should therefore be distinguished from one that predicts labelled subject–relation–object triples in a conventional scene graph.

Kipf and Welling [31] developed a graph convolutional network (GCN) using nor-

malised neighbourhood propagation. Hamilton et al. [32] proposed learning neighbourhood aggregation functions for inductive node representation. Veličković et al. [12] introduced graph attention to learn how neighbours contribute within a supplied connectivity structure. These works established useful aggregation mechanisms, but did not establish that a particular image grouping or feature-distance rule is semantically correct.

The distinction between topology and aggregation is central to GAViT. Selecting a neighbour by cosine similarity answers which representation is eligible to contribute. Attention then determines its contribution according to the graph layer. The two scores need not be identical. A fixed spatial edge can receive a learned attention weight, while a content-selected edge can be used without passing its original similarity value as a message coefficient. Reporting a graph therefore requires specifying both decisions.

Directed edges add another necessary detail. If a query selects a feature neighbour, the intended aggregation usually requires the neighbour's representation to reach the query. Reversing the stored direction can change that operation. The explicit direction, treatment of self-connections and duplicate spatial/feature edges are consequently part of the method definition, even when they are not visible in a schematic diagram.

2.3.2 Dynamic feature graphs and visual backbones

Wang et al. [33] introduced EdgeConv in a dynamic graph network for point clouds. Its feature-space neighbourhoods provide a useful precedent for content-dependent connectivity. However, a point-cloud graph and a regional image graph have different geometric inputs and prediction settings. The precedent supports a design option, not a direct transfer of the reported task performance.

Han et al. [34] represented images through patch nodes in Vision GNN (ViG). Their design combined a graph aggregation module with feature transformations in a visual backbone. Munir et al. [35] subsequently investigated efficient graph construction for vision graph networks. These works connect connectivity choices to computational behaviour: the cost of constructing neighbours is part of an image graph model, rather than a free preprocessing assumption.

Graph recomputation is also a separate architectural choice. A model may construct a feature graph once and reuse it through several message-passing layers, or update neighbours after intermediate features change. Both may be described as content-dependent, but their information paths and costs differ. GAViT uses a low-node-count regional branch above a pretrained visual hierarchy, which differs from replacing that hierarchy with repeated graph blocks. Its evaluation must include the pooling, graph construction

and feedback that create the branch, not only the graph layer in isolation.

2.3.3 Graph and attention as complementary operators

Chen et al. [36] proposed Graph-Based Global Reasoning Networks, which projected coordinate-space features to a smaller interaction space using weighted pooling, performed graph convolution, and broadcast the resulting information back. This is a particularly direct precedent for aggregation, graph reasoning and return to a spatial representation. The distinction of GAViT therefore concerns its anchored grouping, sparse spatial-and-feature topology and late-stage feedback implementation, rather than the general existence of this three-step organisation.

Graph networks and Transformers can be combined at different levels. Rampášek et al. [37] studied a framework that combined local message passing, global attention and structural information for graph learning. This offered a general perspective on complementary information paths, although its graph benchmarks were not a direct evaluation of regional remote sensing image classification.

For visual models, complementary paths can appear as parallel feature branches, alternating blocks or late-stage augmentation. Their meanings depend on the units exchanged between the branches. A channel-attention operation, token self-attention and a graph over pooled regions are not interchangeable merely because all use learned weights. Comparing their insertion points and node granularity is more informative than grouping them under a single attention label.

This observation also bounds claims of novelty. Combining graph learning with attention-based vision has substantial precedent. The contribution of a particular system can instead concern a precisely specified construction, its controlled comparison and what its analysis reveals about the conditions under which the added structure is useful.

2.4 From fixed partitions to learned token groups

2.4.1 Fixed grids and hard feature grouping

Fixed spatial partitions offer stable region identifiers and inexpensive membership rules. They also provide an immediate interpretation for spatial neighbours. Their boundaries are determined by coordinates, however, so a linear structure or mixed scene may cross several groups. Learned weighting within a fixed group changes its pooled representation without changing which tokens belong to it. This distinction is particularly important when the term attention-guided grouping is used.

Hard feature grouping offers a different trade-off. Clustering can place similar tokens together even when their positions differ, but the resulting groups may lose a consistent spatial identity. The choice of feature metric, number of groups, initialisation and handling of empty groups then becomes part of the model. A discrete membership rule also differs from a soft assignment through which downstream gradients can change token membership. These alternatives motivate retaining a fixed grouping as a reference rather than assuming that increased flexibility must improve classification.

2.4.2 Superpixels, locality and differentiable association

Achanta et al. [38] proposed Simple Linear Iterative Clustering (SLIC) superpixels using a local combination of appearance and position. Jampani et al. [39] introduced Superpixel Sampling Networks (SSN), connecting differentiable superpixel formation with learned features and downstream objectives. The shared lesson for this thesis is that content dependence and spatial constraints can coexist. They need not be treated as mutually exclusive grouping strategies.

The level at which grouping operates nevertheless matters. Pixel-level superpixels can be assessed against fine spatial boundaries, whereas a group of late-stage contextual tokens may have a much coarser and overlapping relationship to the input. A fixed anchor can provide a useful prior without being a true object centre. Conversely, a visually irregular group does not automatically imply a poor classification representation. The appropriate claim depends on the supervision, spatial resolution and measured outcome.

Differentiable association also requires specifying its normalisation. A token may distribute membership mass across several regions, while a region separately normalises the weights of all contributing tokens. These answer different questions: where a token belongs, and how a region is represented. Separating the two is useful when the same membership is subsequently used to send region information back to the original tokens.

2.4.3 Object-centric and semantically supervised groups

Locatello et al. [40] introduced Slot Attention to organise input features through a set of competing slots and iterative updates. Its grouping mechanism provides a reference for differentiating input membership from slot aggregation. Its exchangeable slot identities also differ from regions bound to a persistent spatial anchor. A graph defined by neighbouring anchor indices cannot be transferred to arbitrary slots without explaining the new meaning of adjacency.

Xu et al. [41] proposed GroupViT, in which semantic segmentation emerged through image–text supervision and hierarchical visual grouping. It illustrates how the learning signal and grouping architecture interact. GAViT uses scene classification labels rather than the same visual–language supervision, so it cannot inherit GroupViT's semantic segmentation interpretation solely by using learned assignments.

These methods also show why soft grouping should be specified concretely. The term can describe competitive slots, local cluster associations or hierarchical group-token assignments. They need not use the same update rule, locality, forward representation or gradient estimator. A negative result for a single local-prototype design therefore does not reject learnable grouping as a whole; a positive classification result would likewise not establish object discovery without an appropriate evaluation.

2.4.4 Token aggregation, interaction and reconstruction

Wu et al. [42] explored visual Transformers that organised spatial features through visual tokens. Ryoo et al. [43] introduced TokenLearner and TokenFuser in a video representation framework, combining adaptive token extraction with a mechanism for returning information to spatial features. These provide prior examples of reducing the number of units used for interaction while retaining a connection to a denser visual representation.

Huang et al. [44] proposed a vision Transformer with super token sampling, denoted STViT: sparse associations aggregated visual tokens, interaction occurred among super tokens, and information was mapped back. Chang et al. [45] introduced Semantic Token ViT, which uses the same STViT abbreviation for a different architecture. Its STViT-R variant restores spatial representations through attention from image-token queries to semantic-token keys and values. Both studies provide prior examples of compact token interaction and return to a denser spatial representation. GAViT uses its particular regional assignments and sparse graph operator within this established organisational pattern.

Bolya et al. [46] proposed Token Merging to reduce token redundancy through merging. This makes computational efficiency a distinct purpose for grouping. A method that reduces tokens inside a backbone may save subsequent computation, whereas a regional branch added after a full backbone introduces its own cost. Similar reductions in node count therefore do not imply similar overall speed or memory behaviour.

The local prototype variant in this thesis occupies a restricted position among these alternatives. It retains anchor neighbourhoods, constructs prototypes from each image's contextual tokens, and uses one local assignment step before sparse graph reasoning. It

neither implements the complete iterative Slot Attention procedure nor replaces super-token attention with an equivalent operator. Its value is tested as a specified grouping and feedback variant, rather than inferred from the success of a related family.

2.5 Directly related remote sensing graph methods

2.5.1 Hierarchical graph-enhanced Transformers

Li et al. [47] proposed a Hierarchical Graph-Enhanced Transformer Network (HGTNet) for remote sensing scene classification. The method combined hierarchical Transformer representations, attention modules and graph-based feature extraction. It is directly relevant to the claim that explicit relations can complement visual attention in this application. Its placement and organisation of modules differ from a single regional graph above the final stage of a shared Swin-T backbone.

This difference makes HGTNet a complete-method comparator rather than a single-component ablation. A comparison must retain and disclose its backbone, graph operations, feature fusion and pretraining assumptions. Replacing those elements with an available Swin–GAT sequence would create a different implementation. Its published performance also depends on its reported benchmark protocol; it cannot be inserted into a common-protocol result column as though it were measured in this project.

2.5.2 Deep feature aggregation through graph convolution

Xu et al. [48] proposed a Deep Feature Aggregation Framework driven by a Graph Convolutional Network (DFAGCN). Their method combined multiple levels of convolutional features, graph-based aggregation, feature fusion and a final linear classification stage. Its relationship to GAViT lies in aggregating visual information through an explicit graph, while its feature source and processing chain differ from the token-feedback design studied here.

The complete chain matters when defining a comparison. Multiscale feature extraction, graph training, fusion coefficients and the classifier's selection procedure can each change the resulting representation. A standalone graph-convolution layer attached to the present classifier is therefore not a reproduction of DFAGCN. Independent implementation choices must be recorded where the available specification leaves uncertainty. Preparing or verifying individual components also does not establish a completed benchmark result.

Table 2.1 summarises the architectural relationships most relevant to the study. Its entries compare designs, not performance under a shared experimental protocol.

Table 2.1: Selected relational designs and their relationship to GAViT. The table describes architectural organisation rather than a ranking of published classification scores.

Method	Visual units and interaction	Relationship to the present study
GloRe (2019)	Pooled interaction-space graph and broadcasting	Direct graph precedent for aggregation, relational reasoning and mapping back.
ViG (2022)	Patch-node graph backbone	Repeated visual graph processing differs from a late regional branch.
TokenLearner (2021)	Adaptive tokens and spatial fusion	Aggregation and return to spatial features predate the present framework.
Super Token Sampling (2023)	Sparse token association and super-token attention	Shares the broad aggregation–interaction–mapping organisation; the interaction operator differs.
HGTNet (2024)	Hierarchical Transformer and graph feature fusion	Direct remote sensing comparator with a different full architecture.
DFAGCN (2022)	Multilevel CNN features and graph aggregation	Direct graph-based comparator with a different staged training and classification chain.
GAViT	Anchored token groups and sparse regional GAT	Tests fixed and local-prototype membership with a common backbone and feedback interface.

2.6 Evaluation, interpretability and computational trade-offs

Architectural comparisons require compatible evidence. The number of training images, validation rule, pretraining, augmentation and numerical settings can all affect the meaning of a result. Fixed-split training repetitions measure a different source of variation from experiments that redraw the data partition. Published means and standard deviations should therefore be accompanied by their repetition protocol. A higher number in a paper using another split or additional supervision is not an isolated estimate of the benefit of its graph module.

Single-label and multi-label metrics also answer different questions. Overall accuracy counts correct scene decisions, whereas average precision assesses the ranking of examples for a label. Macro-averaged measures weight classes equally, while micro-averaged measures combine their decisions. For a multi-label task, threshold-dependent and ranking metrics can move in different directions. This thesis fixes its principal measures before comparison and preserves per-class outcomes, rather than changing the main criterion after observing favourable results.

Qualitative evidence is useful when its meaning is precise. A partition shows member-

ship, a graph plot shows selected connectivity, and a weight map shows a particular aggregation mechanism. None automatically measures semantic region quality. In particular, similarity-based feature edges can connect predictive contexts without representing named relationships between objects. Evaluating graph reasoning therefore benefits from showing both corrected examples and newly introduced errors, together with full-set classification results. Selected successful images alone cannot establish a general explanation of model behaviour.

Efficiency likewise belongs to the research question. Pooling reduces the number of nodes in a graph but does not eliminate backbone computation or the cost of constructing the graph. Parameter count, forward latency, throughput, memory and total training duration describe different aspects of that cost. Comparisons should state whether inputs are already on the device and whether loading, preprocessing and transfer are included. An added branch can have modest memory overhead yet affect latency through its sequence of operations; this must be measured under shared conditions.

2.7 Synthesis and research position

The literature supports three distinctions that guide the experimental questions in Chapter 1. First, a strong contextual backbone and an explicit regional graph are different forms of organisation. Their combination is plausible, but its incremental benefit must be established through a common-backbone comparison. Second, learnable pooling weights, learnable membership and learned graph attention are separate sources of flexibility. Increasing one does not establish that the resulting regions better represent land-cover entities or improve scene recognition. Third, aggregation and feedback have clear precedents, so the present work is positioned around a defined regional construction and its evaluation.

GAViT accordingly combines a pretrained Swin-T hierarchy with anchored region aggregation, a sparse spatial-and-feature graph, graph attention and feedback to the original tokens. Its fixed and local-prototype variants share the main interfaces while changing the membership mechanism. Chapter 3 specifies the implementation; Chapter 4 separates the data and model-selection protocols needed to assess it.

The intended evidence includes classification performance across tasks and training repetitions, component comparisons, computational cost and diagnostic visualisations. These support a bounded account of when the added structure helps, when it harms, and which explanations remain unresolved. This positioning allows positive and negative observations to inform the same research questions without presupposing that a more elaborate architecture must outperform its shared backbone.

Chapter 3

Graph-Augmented Vision Transformer

3.1 Problem formulation and architectural scope

This study investigates whether explicit region aggregation and graph reasoning provide useful additional structure for remote sensing scene classification. The comparison uses a common Swin-T backbone, so that the principal architectural changes occur after the backbone feature extraction stage. Swin was proposed by Liu et al. [11]; its shifted-window hierarchy already supports contextual interaction. The motivation here is therefore to test an additional regional inductive bias, rather than to assume that the backbone lacks spatial or contextual modelling.

Let $I \in \mathbb{R}^{3\times224\times224}$ denote a preprocessed RGB image, and let C denote the number of output classes. For single-label classification, the target y is one class index in $\{1, \ldots, C\}$. For multi-label classification, the target is a binary vector $\boldsymbol{y} \in \{0, 1\}^C$. The backbone produces a sequence $X = [x_1, \ldots, x_N]^\top \in \mathbb{R}^{N\times D}$, where $N = 49$ and $D = 768$. The tokens occupy a 7×7 spatial grid. They are contextual features from the final backbone stage, rather than independent raw-image patches.

We implement a Graph-Augmented Vision Transformer (GAViT) with four successive operations: token-to-region aggregation, sparse region graph construction, graph attention, and region-to-token feedback. The principal comparison uses either a fixed spatial partition with learned pooling weights or a local, image-dependent soft assignment. Both versions retain $K = 16$ region anchors and the same explicit graph-edge budget. The term *region* denotes a group of feature tokens; it does not imply a supervised object or land-cover segment.

3.2 Shared feature extractor and baseline

The backbone is the `swin_tiny_patch4_window7_224` model. Its final feature map has spatial dimensions 7×7 and 768 channels and is reshaped into X. The experiments fine-tune the backbone from the recorded ImageNet-pretrained checkpoint. No additional classification token is inserted.

The baseline first computes the global representation

$$\bar{x} = \frac{1}{N} \sum_{n=1}^{N} x_n, \tag{3.1}$$

where n indexes tokens. Its logit vector is

$$\boldsymbol{\ell} = W_c \ \mathrm{Dropout}(\mathrm{LN}(\bar{x})) + b_c. \tag{3.2}$$

Here LN denotes layer normalisation, $W_c \in \mathbb{R}^{C \times D}$ and $b_c \in \mathbb{R}^C$ are classifier parameters, and Dropout uses probability 0.1 during training and is disabled during evaluation. GAViT uses the same form of classifier after its feedback operation.

3.3 Fixed spatial partition with attentive pooling

3.3.1 Deterministic token membership

For the assignment formula only, region identifiers are zero-based. A token at grid coordinate (u, v), with $u, v \in \{0, \ldots, 6\}$, is assigned to

$$a(u, v) = 4 \left\lfloor \frac{4u}{7} \right\rfloor + \left\lfloor \frac{4v}{7} \right\rfloor. \tag{3.3}$$

The function a maps each token to one of the 16 anchors. We write $a(n)$ for the assignment of token n. Along each grid axis, the resulting group sizes are $[2, 2, 2, 1]$. Thus, nine regions contain four tokens, six contain two tokens, and one contains one token. This is a partition of a 7×7 feature grid; it is not a division into 16 equally sized input-image crops.

3.3.2 Learned pooling within each region

A shared scorer produces one scalar $s_n = f_\theta(x_n)$ per token. The scorer f_θ consists of a 768-to-192 linear layer, a Gaussian error linear unit (GELU), and a 192-to-1 linear layer. Its trainable parameters are denoted collectively by θ. The fixed-membership pooling weights are

$$w_{nk}^{(0)} = \frac{\mathbf{1}[a(n) = k] \exp(s_n)}{\sum_{m=1}^{N} \mathbf{1}[a(m) = k] \exp(s_m)}, \qquad r_k = \sum_{n=1}^{N} w_{nk}^{(0)} x_n, \tag{3.4}$$

where $k \in \{0, \ldots, K-1\}$ indexes regions, m is a second token index, $\mathbf{1}[\cdot]$ is an indicator, and $r_k \in \mathbb{R}^D$ is the pooled region feature. The weights sum to one over the tokens in each region. The scorer changes which member tokens contribute most strongly, but it does not move the partition boundaries or change the member set.

3.4 Local prototype soft assignment

The soft version changes token membership while preserving the anchor layout. It constructs image-dependent prototypes from the fixed partition, measures token–prototype similarity in a learned projection space, and restricts assignment to nearby anchors. This is a single assignment step; it is not an iterative clustering procedure or a bank of independently learned class prototypes.

3.4.1 Projected tokens and image-dependent prototypes

Let $P \in \mathbb{R}^{128\times D}$ denote a learned linear projection without bias. Define

$$z_n = \mathrm{unit}(P\,\mathrm{LN}_a(x_n)), \qquad p_k = \mathrm{unit}\left(\sum_{n=1}^{N} w_{nk}^{(0)} z_n\right), \tag{3.5}$$

where LN_a is the assignment projection's layer normalisation, $z_n \in \mathbb{R}^{128}$ is a normalised token projection, and $p_k \in \mathbb{R}^{128}$ is the current image's prototype for anchor k. The operation $\mathrm{unit}(v) = v/\max(\|v\|_2, \epsilon)$ normalises a vector v, with a small positive numerical stabiliser ϵ. The initial weights $w^{(0)}$ are those in Equation (3.4). Consequently, the prototypes change with the image and the backbone features.

3.4.2 Local membership and separate aggregation weights

Let $g(k) = (\lfloor k/4 \rfloor, k \bmod 4)$ be the grid coordinate of anchor k. The candidate set for token n is

$$C(n) = \{k : \|g(a(n)) - g(k)\|_\infty \le 1\}, \tag{3.6}$$

where $\|\cdot\|_\infty$ measures the largest absolute coordinate difference. This restriction permits assignment to adjacent anchors, including diagonal neighbours, while maintaining a spatial prior. With temperature $\tau = 0.1$, the membership matrix $S \in \mathbb{R}^{N\times K}$ is

$$S_{nk} = \begin{cases} \dfrac{\exp(z_n^\top p_k/\tau)}{\sum_{j\in C(n)} \exp(z_n^\top p_j/\tau)}, & k \in C(n), \\ 0, & k \notin C(n). \end{cases} \tag{3.7}$$

Here j indexes candidate regions, and $\sum_k S_{nk} = 1$ for every token.

Membership and region pooling use different normalisations. Define the pooling matrix $W \in \mathbb{R}^{N\times K}$ by

$$W_{nk} = \frac{S_{nk}\exp(s_n)}{\sum_{m=1}^{N} S_{mk}\exp(s_m)}, \qquad r_k = \sum_{n=1}^{N} W_{nk} x_n. \tag{3.8}$$

Thus, S sums to one over regions for each token, whereas W sums to one over tokens for each region. The implementation computes the latter operation using $\log S$ and a token-axis softmax. Region aggregation uses the original 768-dimensional tokens, rather than

the 128-dimensional projection.

The projection and assignment normalisation add 99,840 trainable parameters relative to the fixed grouping module. Gradients pass through the local membership calculation, the image-dependent prototypes and the pooling weights. The locality mask remains fixed. The resulting regions may overlap in their token contributions, but they retain anchor identities and are not unconstrained semantic segments.

3.5 Sparse spatial and feature graph

The graph is constructed once per image from its pooled region features. Its node set is $\mathcal{V} = \{0, \ldots, K-1\}$. Let $\mathcal{N}_4(i)$ denote the up, down, left and right anchor neighbours of node i, restricted to the 4×4 grid. The directed spatial edge set is

$$\mathcal{E}_s = \{(j \to i) : j \in \mathcal{N}_4(i)\}. \tag{3.9}$$

It contains 48 edges. Each neighbouring anchor pair contributes both message directions.

Feature edges provide a second source of connectivity. For every query node i, the cosine score to candidate node j is

$$c_{ij} = \frac{r_i^\top r_j}{\max(\|r_i\|_2, \epsilon)\max(\|r_j\|_2, \epsilon)}. \tag{3.10}$$

Let $\mathcal{T}_2(i)$ contain the two candidates with the highest scores among $\mathcal{V} \setminus (\{i\} \cup \mathcal{N}_4(i))$. Equal scores are resolved by preferring the smaller region index. The feature and combined edge sets are

$$\mathcal{E}_f = \{(j \to i) : j \in \mathcal{T}_2(i)\}, \qquad \mathcal{E} = \mathcal{E}_s \cup \mathcal{E}_f. \tag{3.11}$$

There are 32 feature edges and 80 unique explicit non-self edges in total. The count describes the graph supplied to the graph-attention operator, before its internal edge processing. It is not a count of every message handled internally by that operator.

An edge $j \to i$ supplies information from selected neighbour j to query i. The feature graph depends on the image, while the spatial graph retains the fixed anchor adjacency in both grouping variants. In particular, soft region geometry is not used to recompute spatial touching relationships or region centroids. Cosine similarity selects connectivity; its numeric value is not supplied as an additional message weight to the graph-attention layers. The top-two selection is discrete, so differentiable soft membership does not make the entire graph topology differentiable. Both graph-attention layers share the graph constructed before message passing.

3.6 Graph attention and token feedback

Graph attention networks (GATs) were proposed by Veličković et al. [12] to learn neighbour aggregation weights. Here, let $R \in \mathbb{R}^{K\times D}$ be the matrix of region features. An input linear projection maps its rows to width 1024, yielding $H^{(0)} \in \mathbb{R}^{K\times 1024}$. Two graph-attention layers then apply

$$H^{(l+1)} = \mathrm{Dropout}\Big(\mathrm{GELU}\Big[\mathrm{LN}_l\Big(\mathrm{GAT}_l(H^{(l)}, \mathcal{E}) + H^{(l)}\Big)\Big]\Big), \quad l \in \{0, 1\}. \tag{3.12}$$

The layer index is l, LN_l is its layer normalisation, and GAT_l has four concatenated attention heads with 256 output channels per head. Both attention dropout and the output dropout use probability 0.1. The residual connection preserves the 1024-dimensional shape.

Let $h_k^{(2)}$ denote row k of $H^{(2)}$. A learned projection $Q \in \mathbb{R}^{D\times 1024}$ with bias $b_f \in \mathbb{R}^D$ and a further layer normalisation LN_f produces

$$F_k = \mathrm{LN}_f(Q h_k^{(2)} + b_f), \tag{3.13}$$

where $F_k \in \mathbb{R}^D$ is the feedback for region k. The updated tokens are

$$x'_n = \begin{cases} x_n + F_{a(n)}, & \text{fixed grouping,} \\ x_n + \sum_{k=0}^{K-1} S_{nk} F_k, & \text{soft grouping.} \end{cases} \tag{3.14}$$

No learned scalar gate is applied at this interface. The updated tokens are mean-pooled and passed to the classifier in Equation (3.2). The principal model does not apply another Transformer block after feedback.

This placement also limits the interpretation of token feedback. In the soft version, the vector passed to the classifier satisfies

$$\frac{1}{N}\sum_{n=1}^{N} x'_n = \bar{x} + \sum_{k=0}^{K-1} \mu_k F_k, \qquad \mu_k = \frac{1}{N}\sum_{n=1}^{N} S_{nk}. \tag{3.15}$$

Here μ_k is the average membership mass of region k. This identity describes the final pooling step, not the complete effect of assignment: S also changes the upstream region features, feature-edge selection and graph outputs. It nevertheless prevents an unsupported claim that feedback is followed by further token-to-token reasoning in the principal model.

3.7 Classification objectives

For single-label tasks, let q_c be the softmax probability of class c, computed from logit ℓ_c. The per-image cross-entropy objective is

$$q_c = \frac{\exp(\ell_c)}{\sum_{j=1}^{C} \exp(\ell_j)}, \qquad \mathcal{L}_{\text{single}} = -\log q_y, \tag{3.16}$$

where j now indexes output classes and y is the target class. For multi-label tasks, define $p_c = \sigma(\ell_c)$, where σ is the sigmoid function and y_c is the binary target for class c. The objective is

$$\mathcal{L}_{\text{multi}} = -\frac{1}{C} \sum_{c=1}^{C} \left[y_c \log p_c + (1 - y_c) \log(1 - p_c) \right]. \tag{3.17}$$

Both objectives are averaged over the training minibatch. The implementation uses binary cross-entropy directly on logits for numerical stability in the multi-label case. No region masks, object annotations or auxiliary grouping loss are used in the principal models. Consequently, classification training alone does not establish that the learned grouping recovers semantic boundaries. That question requires separate evidence from the classification comparison and qualitative analysis.

Chapter 4

Experimental Setup

4.1 Evaluation questions

The experiments address three questions. First, does the fixed region graph improve classification relative to the shared Swin-T baseline? Second, does the local prototype soft assignment improve on the fixed grouping under the same task protocol? Third, are any observed benefits consistent across training repetitions and sufficient to justify the additional computational cost? Classification results, component comparisons and diagnostic visualisations provide different kinds of evidence for these questions.

This chapter specifies the implemented internal comparison. Chapter 5 separately reports the completed single-seed independent implementations of HGTNet and DFAGCN, including their different preprocessing, optimisers and training budgets. The shared partitions and evaluation rules do not make these separate recipes a unified training protocol. Published scores are not treated as measurements made under the internal protocol; external multi-seed and unified-protocol comparisons remain open.

4.2 Datasets and fixed partitions

4.2.1 NWPU-RESISC45

Cheng et al. [3] introduced NWPU-RESISC45 as a remote sensing scene classification benchmark with 45 classes and 31,500 images, including 700 images per class. The present experiments use a 20% training pool and an 80% held-out test set. The training pool is divided into a calibration-training subset and a validation subset before final fitting. Table 4.1 gives the retained image counts.

The outer partition seed is 42 and the within-pool validation partition seed is 4242. These fixed partition seeds are distinct from the seeds used for training repetitions. The manifest construction identified eight groups of same-class images with identical pixel content. Each such group is kept within one subset. All original files are retained, so the manifest contains 31,500 files representing 31,492 unique pixel images. This check addresses the identified exact duplicates; it does not establish the absence of near-duplicate images or geographic dependence.

Table 4.1: Retained sample counts for the implemented internal protocols. "Pool" is the union of calibration training and validation for the single-label datasets. BigEarthNet keeps training and validation separate.

Dataset	Training	Validation	Pool	Test
NWPU-RESISC45	5,040	1,260	6,300	25,200
AID	4,000	1,000	5,000	5,000
BigEarthNet-19	237,871	122,342	—	119,825

4.2.2 Aerial Image Dataset

The Aerial Image Dataset (AID) was introduced by Xia et al. [4] for aerial scene classification. It contains 10,000 images from 30 scene classes. The implemented partition uses a 50% training pool and a 50% held-out test set. The pool is divided into 4,000 calibration images and 1,000 validation images. The outer and within-pool partition seeds are again 42 and 4242. One same-class exact-duplicate group is confined to one subset; all 10,000 files are retained, with 9,999 unique pixel images.

4.2.3 BigEarthNet-19

Sumbul et al. [5] introduced BigEarthNet as a multi-label Sentinel-2 archive. The 19-class nomenclature was described by Sumbul et al. [6]. These dataset references do not define the number of samples retained by this project's available-data pipeline. The implemented manifests use the existing partition labels in BigEarthNet v2.0's `metadata.parquet` [49] and retain patches with the required RGB files. They contain 480,038 samples in total, divided as shown in Table 4.1; they are not a newly generated random 70/15/15 split.

Only Sentinel-2 bands B04, B03 and B02 are used, in red, green and blue order. When separate band files are read, each band value is divided by 4095, multiplied by 255, clipped to $[0, 255]$ and converted to an unsigned 8-bit image. The reader also supports previously merged RGB images. The inputs are therefore three-channel RGB representations, rather than full multispectral or Sentinel-1/Sentinel-2 inputs. Targets use a fixed ordering of 19 binary land-cover labels.

4.3 Preprocessing and optimisation

All internal Swin-T and GAViT models receive 224×224 RGB inputs. Their evaluation uses deterministic resizing and ImageNet channel normalisation, with means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. The single-label pipeline uses bilinear resizing with antialiasing. During single-label training, horizon-

tal and vertical flips each have probability 0.5; brightness, contrast and saturation jitter each use magnitude 0.2. BigEarthNet training uses the resize and two flips, without colour jitter. Random training augmentations are disabled during validation and testing.

The internal comparison uses minibatches of 32 and the AdamW optimiser, which was proposed by Loshchilov and Hutter [50]. The initial learning rate is 3×10^{-4} and weight decay is 10^{-4}. A cosine schedule uses an annealing horizon $T_{\max} = 30$ epochs and a minimum learning rate of zero. The single-label pipeline uses AdamW moment coefficients $(0.9, 0.999)$ and numerical stabiliser 10^{-8}. It applies cross-entropy without label smoothing. BigEarthNet uses binary cross-entropy with logits. The backbone is fine-tuned together with the classification and graph modules.

The recorded principal internal runs use 32-bit floating-point computation without automatic mixed precision (AMP). TensorFloat-32 (TF32) settings are recorded separately: the single-label pipeline disables TF32 for matrix multiplication and cuDNN, whereas the documented BigEarthNet soft-seed42 environment disables matrix-multiplication TF32 and retains cuDNN TF32. These settings describe the respective pipelines; they do not establish that every historical run used an identical software or hardware environment.

4.4 Model selection and final fitting

4.4.1 Single-label calibration and refitting

For AID and NWPU-RESISC45, each internal Swin-T or GAViT model first trains for at most 30 calibration epochs on the internal training subset. Validation overall accuracy selects the epoch count. The checkpoint is updated only on a strict improvement, so the earliest occurrence is retained when validation accuracy ties.

After selection, the model is reinitialised from the common pretrained backbone checkpoint and fitted on the full training pool for the selected number of epochs. This refit is not continued training from the calibration checkpoint. The cosine schedule retains its annealing horizon $T_{\max} = 30$ epochs; it is not rescaled to the selected refit duration. The final refit checkpoint is evaluated on the held-out test set. Thus, a calibration validation score and a refit test score can come from different trained weights. A reversal between their rankings alone does not demonstrate overfitting of a single checkpoint.

4.4.2 Multi-label validation-based checkpoint selection

BigEarthNet uses its training partition for optimisation and its validation partition for checkpoint selection. The planned training budget is 30 epochs, and the best checkpoint is updated when validation mean average precision (mAP) strictly improves. The selected checkpoint is then evaluated on the fixed test partition. Training and validation are not merged for refitting. The achieved final epoch and exit status are reported per run, since some historical runs ended before the planned budget.

The F1 decision threshold is fixed at 0.5. Neither checkpoint choice nor threshold choice uses the test labels. A higher test F1 score does not replace mAP as the preselected primary multi-label metric.

4.5 Classification metrics

4.5.1 Single-label metrics

Let M be the number of test images, and let y_i and $\hat{y}_i$ be the true and predicted class of image i. The predicted class is the largest-logit class. Overall accuracy (OA) is

$$\mathrm{OA} = \frac{1}{M} \sum_{i=1}^{M} \mathbf{1}[\hat{y}_i = y_i]. \tag{4.1}$$

For class c, let TP_c, FP_c and FN_c denote true positives, false positives and false negatives, respectively. The per-class and macro F1 scores are

$$F1_c = \frac{2TP_c}{2TP_c + FP_c + FN_c}, \qquad F1_{\mathrm{macro}} = \frac{1}{C} \sum_{c=1}^{C} F1_c. \tag{4.2}$$

A zero denominator gives an F1 score of zero in the implemented metric. Per-class accuracy is $TP_c/(TP_c + FN_c)$ when the class has test support. The confusion matrix uses true classes as rows and predicted classes as columns. Scores are displayed as percentages in result tables.

4.5.2 Multi-label metrics

Sigmoid probabilities rank the test images independently for each label. Let AP_c denote class c's average precision, calculated from that ranking using the recorded evaluation implementation. For C_+ classes with at least one positive test example, mAP is

$$\mathrm{mAP} = \frac{1}{C_+} \sum_{c:\, TP_c + FN_c > 0} AP_c. \tag{4.3}$$

The intended full BigEarthNet-19 evaluation covers all 19 labels; absent classes in small engineering subsets are not grounds for treating subset mAP as a benchmark result. The class-wise AP vector is retained alongside the aggregate score.

Binary predictions use $\hat{y}_{ic} = \mathbf{1}[p_{ic} \geq 0.5]$, where p_{ic} is image i's sigmoid probability for class c. Macro-F1 uses Equation (4.2). Micro-F1 aggregates counts over all labels:

$$F1_{\text{micro}} = \frac{2\sum_c TP_c}{2\sum_c TP_c + \sum_c FP_c + \sum_c FN_c}. \tag{4.4}$$

Average precision measures ranking, whereas F1 also depends on the fixed decision threshold. Their changes can therefore have different directions.

4.6 Internal comparisons and repetition statistics

The principal architecture comparison contains the shared Swin-T baseline, fixed-partition GAViT, and local-prototype soft-assignment GAViT. Within a dataset, comparisons retain the same recorded data partitions, label order, input representation and model-selection rule. The soft variant changes both region aggregation and the feedback assignment. Accordingly, its comparison with the fixed variant does not separately isolate the effects of membership softness, the projection parameters and the feedback map.

Let v_{ms} be the test metric of method m in training repetition s, and let R be the number of eligible repetitions in a stated comparison group. The reported mean $\bar{v}_m$ and sample standard deviation s_m are

$$\bar{v}_m = \frac{1}{R}\sum_{s=1}^{R} v_{ms}, \qquad s_m = \sqrt{\frac{1}{R-1}\sum_{s=1}^{R}(v_{ms} - \bar{v}_m)^2}, \quad R > 1. \tag{4.5}$$

No standard deviation is assigned to a single run. A paired difference is computed within the same seed before its mean and standard deviation are calculated. Differences between percentage scores are expressed in percentage points (pp).

Each result table states its actual seed set and completion criteria. Historical runs with different code identities, devices, training durations or evidence completeness are disclosed separately. They are not relabelled as fully controlled repetitions. All prespecified runs, including negative results and interrupted runs, remain in the experiment record. A subset prepared for a progress report does not replace the complete experiment record or justify discarding unfavourable runs.

Training repetitions share the frozen test set. They measure variation conditional on that partition and do not create additional independent test images or reproduce a repeated-random-split protocol. Means and standard deviations alone do not establish statistical

significance or equivalence. No significance-test outcome is asserted in this partial draft; the final test specification must match the completed paired comparisons and their available sample size.

4.7 Efficiency and qualitative analysis

Efficiency reporting separates parameter count, model-forward latency, throughput, GPU memory and total experiment duration. A model-forward benchmark uses the same device, input shape, batch size, numerical precision, warm-up and synchronisation procedure for all compared models. The archived AID forward benchmark uses GPU-resident synthetic inputs, 20 warm-up passes and five groups of 50 timed passes. It excludes image reading, preprocessing and host-to-device transfer. Its throughput therefore describes the measured forward path, rather than the complete dataset pipeline. Repeated timing groups are not independent training seeds, and runs made on different GPU models are not direct evidence of a model's relative execution speed.

Qualitative analysis distinguishes region assignment, explicit spatial and feature edges, within-region pooling weights, and graph-attention weights. These visual quantities represent different operations. Feature similarity and graph attention are not interchangeable, and neither alone proves a human-semantic relationship or a causal source of classification improvement. Anchor locations are distinguished from region centroids. Successes, introduced errors and shared failures are retained within each declared selection set. The visualised split and checkpoint identity accompany the figures, since calibration examples cannot substitute for refit test performance.

4.8 Reproducibility and reporting boundaries

Run records bind the data manifest, pretrained weight identity, model configuration, source revision, training seed, numerical settings and checkpoint to their logs and evaluation outputs. Best validation epoch and last achieved epoch are recorded separately. A model option in the current source code is not, by itself, evidence that the corresponding formal experiment was completed.

Earlier NWPU experiments used a different data split, and earlier graph variants used different connectivity or integration rules. Those results are not pooled with the present 20%-training-pool refit protocol. Likewise, the historical 3×3 BigEarthNet experiment uses a separately frozen configuration; it does not change the $K = 16$ definition of the principal model in Chapter 3. The supplementary external-method runs in Chapter 5 use their own disclosed implementation and recipe identities. Their completion does not establish official reproduction or complete the external multi-seed requirement. Fi-

nal eligible seed groups, inferential statistics and complete cross-model efficiency tables must be resolved against their own evidence before the thesis makes a final comparative claim.

Chapter 5
Supplementary External-Method Comparisons

5.1 Scope and evidence

The internal Swin-T comparison tests the effect of adding the implemented regional graph to a shared backbone. A complementary question is how that design compares with complete published graph-based scene-classification methods. HGTNet [47] and DFAGCN [48], reviewed in Chapter 2, provide two relevant architectural comparators. This chapter reports three completed measurements: HGTNet on NWPU-RESISC45 and AID, and DFAGCN on NWPU-RESISC45. Each measurement uses training seed 42. The implementations are independent project versions with explicit choices where the available method specification was incomplete; they are not the authors' official implementations or verified reproductions of the published scores.

The runs reuse the frozen partitions in Table 4.1. NWPU-RESISC45 uses 5,040 calibration-training images, 1,260 validation images, a 6,300-image final fitting pool and 25,200 test images. AID uses 4,000/1,000/5,000/5,000 images, respectively. Class order and sample identity are fixed by the original manifests. Validation selects the epoch count and, for DFAGCN, the linear-classifier parameter. The learned calibration state is then discarded and final fitting starts afresh on the training pool. Test predictions are used for final reporting and subsequent error description, rather than recipe selection.

Original external-method test predictions have been recovered and their aggregate metrics, class metrics and confusion matrices independently recomputed locally. The HGTNet calibration and refit histories are also available. The original AID Swin-T and fixed-GAViT predictions have also been recovered, matched to all 5,000 ordered test paths and labels, and recomputed. The internal NWPU reference scores retain their archived seed42 summaries; the corresponding raw vectors were not recovered in this analysis. These tables are supplementary single-seed complete-method comparisons. They neither replace the internal multi-seed results and component ablations nor complete the required external multi-seed comparison under a unified training protocol.

Table 5.1: Principal recipe differences in the supplementary seed42 comparisons. The common partition does not imply a common training recipe.

Method	Representation and input	Optimisation and selection
Internal Swin-T / GAViT	Shared pretrained Swin-T; RGB resized to 224×224.	Batch 32; AdamW, 3×10^{-4} learning rate, 10^{-4} weight decay; 30-epoch cosine calibration, followed by fresh refit for the selected epoch count.
Independent HGTNet	Partial torchvision Swin-B initialisation; resize to 256×256, then a 224×224 crop.	Batch 16; Adam, 10^{-5} learning rate, 5×10^{-5} weight decay; 100 calibration epochs, no scheduler, followed by fresh refit for the validation-selected epoch count.
Independent DFAGCN	Frozen torchvision VGG16; 224×224 fully connected features and 448×448 convolutional features; three graph branches and weighted fusion.	Graph-branch Adam training, batch 100, initial learning rate 10^{-3}; 100-epoch calibration cap and a factor 0.1 decay every 50 epochs; validation-only epoch and LIBLINEAR C selection before fresh refit.

5.2 Disclosed differences between recipes

The methods share data partitions, label mappings and the separation of selection, final fitting and testing. Their backbones, preprocessing, optimisation and calibration budgets differ. Table 5.1 summarises the principal differences. Consequently, a score difference cannot be attributed solely to the presence or type of graph reasoning.

For HGTNet, training uses a random crop after resizing and a horizontal flip with probability 0.5; evaluation uses a centre crop. Both paths use ImageNet channel normalisation. Computation is FP32 without AMP or TF32. The crop details, optimiser defaults, absence of scheduling and selection rules are disclosed engineering choices rather than asserted original-paper defaults. Partial Swin-B initialisation does not establish equivalence to the original method's pretrained model.

DFAGCN uses a separate staged pipeline, not the end-to-end Swin/GAViT trainer. VGG16 [14] features are frozen, while graph branches are fitted on the permitted training subset. The independent implementation uses 7×7 adaptive pooling, two 256-wide graph convolutional layers per branch and explicit readout and fusion choices. Its torchvision weights and RGB stretch resizing do not establish numerical equivalence to the original MatConvNet preprocessing. The selected final classifier uses LIBLINEAR with $C = 10$, after an independently initialised 51-epoch graph refit. These disclosed choices remain part of the result's identity.

Table 5.2: Supplementary NWPU-RESISC45 comparison at training seed 42. Scores are percentages; different complete-method recipes are disclosed in Table 5.1. No single-run standard deviation is assigned.

Model	OA	Macro-F1	Correct	Refit epochs
Swin-T	93.9325	93.9091	23,671	29
GAViT, fixed 4×4	94.1032	94.0899	23,714	29
GAViT, local soft	93.8690	93.8550	23,655	28
DFAGCN, independent	86.5317	86.4589	21,806	51
HGTNet, independent	94.1190	94.1053	23,718	95

5.3 Single-seed test measurements

5.3.1 NWPU-RESISC45

Table 5.2 reports the complete 25,200-image test results. The external rows are local reconstructions from returned predictions; the internal rows retain their archived seed 42 values. All scores refer to final fitting on the 6,300-image pool rather than calibration validation.

HGTNet exceeds fixed GAViT by approximately 0.015873 pp OA and 0.015358 pp macro-F1, corresponding to four additional correct predictions in aggregate. This is a close single-seed point estimate, rather than evidence of statistical equivalence or a stable ranking. The aggregate difference of four does not reveal how many individual errors either method corrects or introduces; that requires both original prediction vectors.

Fixed GAViT exceeds the independent DFAGCN result by approximately 7.571429 pp OA. This compares a fine-tuned Transformer with a frozen-VGG16, staged graph-and-linear-classifier pipeline under their disclosed recipes. It does not isolate a graph operator or establish that the published DFAGCN method has been reproduced. In contrast, the two returned external prediction vectors can be paired: HGTNet corrects 2,415 DFAGCN errors and introduces 503 errors relative to DFAGCN, a net increase of 1,912 correct predictions. Both directions are retained; this remains a complete-method comparison.

5.3.2 AID

The AID comparison in Table 5.3 uses the original 5,000-image held-out test partition. HGTNet correctly classifies 4,846 images, leaving 154 errors. Its locally reconstructed macro-F1 is 96.77286814931982%. The internal fixed GAViT result is ahead by one correct prediction: 0.02 pp OA and approximately 0.008259 pp macro-F1. HGTNet is

Table 5.3: Supplementary AID comparison at training seed 42, with 5,000 test images. All three rows are independently recomputed from recovered original predictions; their training recipes remain different.

Model	OA	Macro-F1	Correct	Refit epochs
Swin-T	96.34	96.2209	4,817	21
GAViT, fixed 4×4	96.94	96.7811	4,847	20
HGTNet, independent	96.92	96.7729	4,846	96

Table 5.4: Paired AID test outcomes with HGTNet as the candidate. "Corrects" and "introduces" refer to errors relative to the named reference. Both outcome directions are retained.

Reference	Both correct	Corrects	Introduces	Both wrong	Net
Swin-T	4,738	108	79	75	+29
Fixed GAViT	4,771	75	76	78	−1

0.58 pp OA above the archived Swin-T seed42 result. These are numerical differences for the stated runs; they do not establish multi-seed superiority.

The one-image difference masks changes in both directions. Table 5.4 uses the verified original prediction vectors. Relative to fixed GAViT, HGTNet corrects 75 errors but introduces 76, with 78 errors remaining common to both models. Thus 151 images have discordant correctness outcomes despite a net difference of one correct image. Relative to Swin-T, HGTNet corrects 108 errors and introduces 79. These are descriptions of this fixed test partition and trained seed; they do not establish a stable ranking or isolate the graph design from the different calibration budgets, pretraining and augmentation.

No DFAGCN AID measurement is claimed here.

5.4 AID selection history and error structure

HGTNet completes all 100 calibration epochs. The validation rule retains the earliest strict maximum in correct predictions, selecting epoch 96 with 972/1,000 correct: OA97.20% and macro-F1 97.074557%. This maximum occurs only once. The final calibration epoch 100 has 971/1,000 correct, OA97.10% and macro-F1 96.990538%. Figure 5.1 retains the complete calibration history and the subsequent fresh 96-epoch refit, including both the selected and final calibration epochs.

The refit is reinitialised from the same initial model state as calibration, rather than continued from its best checkpoint. An interrupted refit attempt is resumed from the last committed epoch 75 state. The original partial updates are retained in the attempt record but excluded from the accepted trajectory, which finishes with 55,048 optimiser steps.

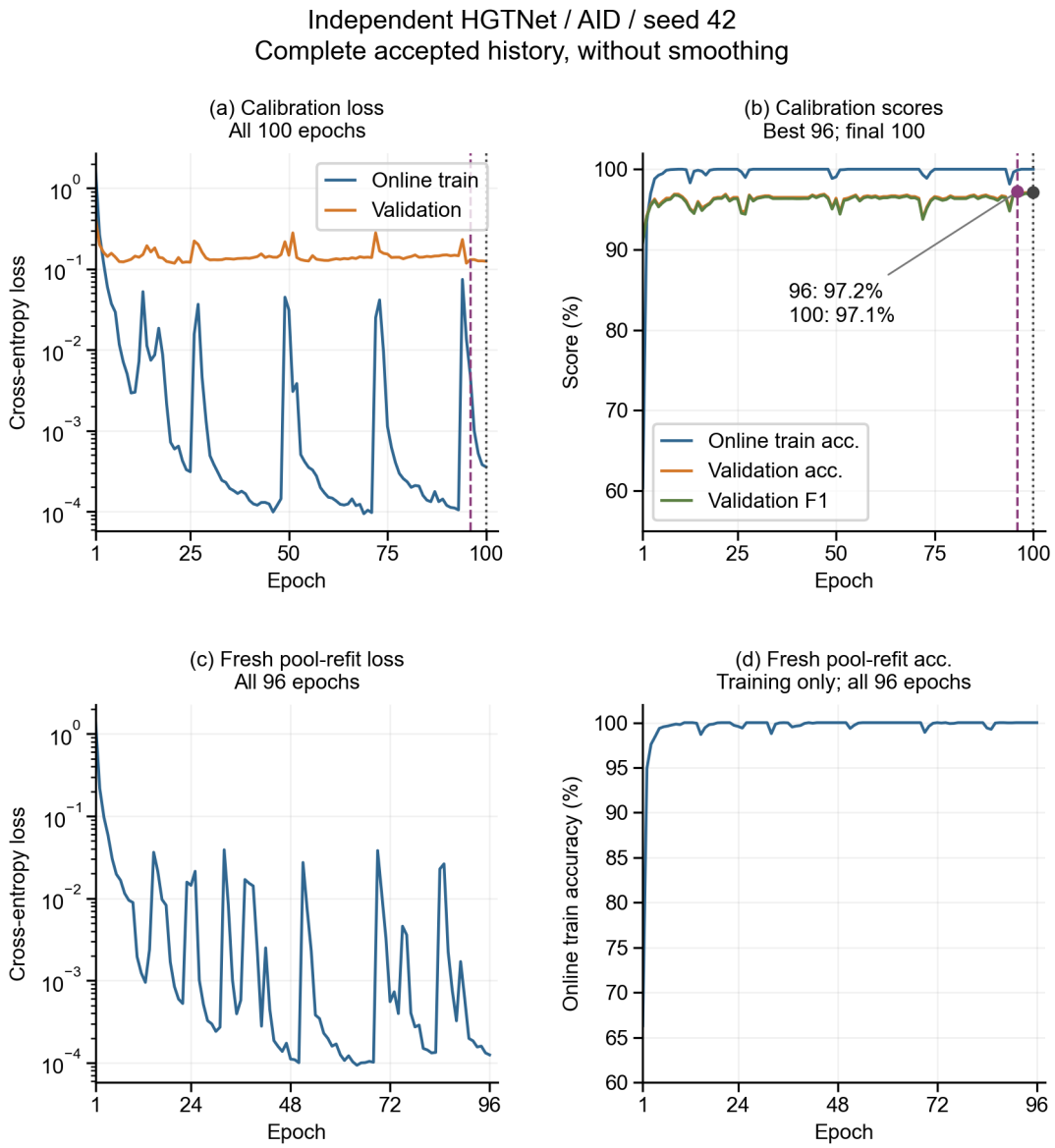


Figure 5.1: Complete HGTNet AID seed 42 calibration and fresh-refit histories. Calibration runs for 100 epochs and selects epoch 96 using validation only; refit starts from the same initial state and runs for 96 epochs on the 5,000-image pool. Refit training accuracy is not a held-out test metric.

The final checkpoint is reloaded with zero maximum discrepancy in the recorded reload check, and the final test transaction completes. This execution evidence supports the identity of the reported run; it is not an additional training seed. The reload check was executed on the server, while local verification recomputes metrics from downloaded records rather than loading the model binary.

Figure 5.2 shows all 30 classes. The lowest per-class recalls are Square 143/165 (86.67%), Center 113/130 (86.92%), Resort 129/145 (88.97%) and School 135/150 (90.00%). Two largest bidirectional confusions tie at 12 errors each: Center/Square contributes 6 in each direction, while Park/Resort contributes 4 Park-to-Resort and 8 Resort-to-Park errors. These counts describe the retained test errors. Without a controlled module intervention or image-level explanatory evidence, they do not identify graph propagation or a particular visual structure as the cause.

5.5 Complexity and remaining comparison requirements

The full neural parameter count is 31,375,385 for fixed GAViT on AID and 74,612,163 for the independent HGTNet. On NWPU-RESISC45 the corresponding counts are 31,386,920 and 74,619,858. Fixed GAViT therefore uses approximately 57.95% fewer parameters on AID and 57.94% fewer on NWPU, while the OA point estimates are close in these two seed 42 comparisons. This is a statement about parameter counts and observed scores, rather than proof that its extra graph computation is worthwhile relative to Swin-T. Relative to the 27,543,960-parameter AID Swin-T baseline, fixed GAViT adds approximately 13.91% parameters. Its smaller size relative to HGTNet therefore does not remove the need to justify its extra cost over its own shared backbone.

The internal seed 42 runs use an RTX 2080 Ti, the DFAGCN NWPU run an RTX 3080, and the HGTNet NWPU run an RTX 3090. The recovered HGTNet AID result resumes on an RTX 3060. Its resumed attempt takes 4,986.5883 seconds and records 4.406GiB peak allocated and 4.820GiB peak reserved CUDA memory. These measurements cover that invocation, including its diagnostics and reload, rather than the whole experiment or inference alone. Full experiment cost and peak memory across attempts remain unavailable. Device, timing scope, preprocessing and training-budget differences prevent a same-device speed or cost claim from these records.

The supported observation is that the independent HGTNet and fixed GAViT have close accuracy point estimates on both held-out datasets for seed 42, with substantially different parameter counts and training recipes. The independent DFAGCN recipe is lower on NWPU in this run. Neither observation establishes a stable complete-method ranking, a causal benefit of a graph component, or equivalence to author-reported results.

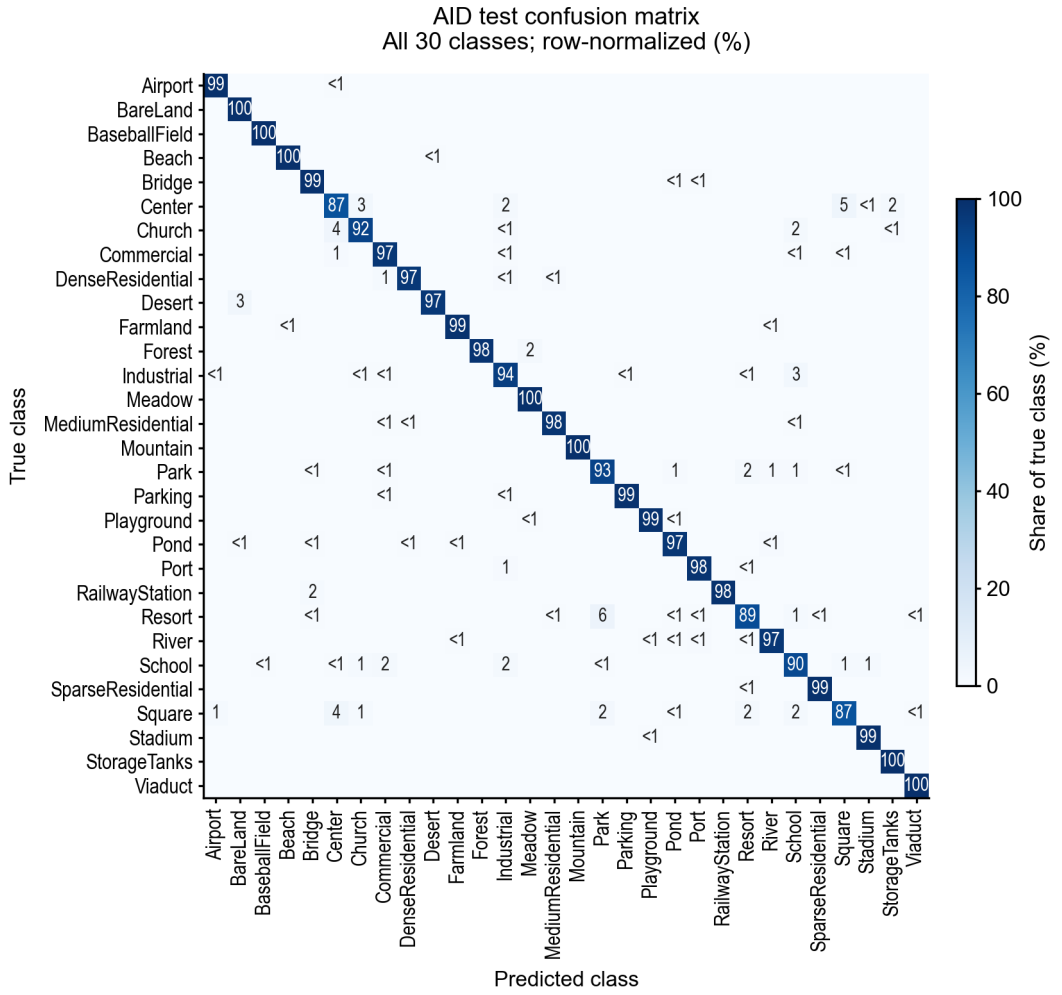


Figure 5.2: HGTNet AID seed 42 confusion matrix for all 5,000 test images. Rows are true classes and columns are predictions; each row is normalised by its class support. All 30 classes are retained, including classes with no observed errors. Colours use unrounded row percentages. Nonzero labels are rounded to whole percentages; values below 1% are shown as < 1. Exact counts are retained in the accompanying records.

A final external main comparison still requires an explicitly unified protocol, multiple completed training seeds with means and sample standard deviations, and comparable efficiency measurements. The complete internal results, architecture ablations and integrated discussion also remain to be added to this partial thesis draft.

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